Stock Market Prediction using LSTM

... (Based on Google Stock Data Analysis &Forecasting)

1. Introduction

Stock market prediction has long been a challenging yet highly valuable problem in the field of finance and data science. Accurate prediction of stock prices can assist investors, traders, and financial institutions in making informed decisions. However, due to the highly volatile and nonlinear nature of financial markets, traditional statistical models often fail to capture complex patterns.

With the advancement of deep learning, models such as Long Short-Term Memory (LSTM) networks have shown promising results in time-series forecasting tasks. LSTM, a special type of Recurrent Neural Network (RNN), is particularly effective in capturing long-term dependencies in sequential data such as stock prices.

In this project, we aim to build a stock price prediction system using historical data of Google (GOOG) stock. The model leverages LSTM to learn patterns from past price movements and predict future trends. Additionally, moving averages and visualization techniques are used to analyze trends before applying machine learning.

2. Methods

2.1 Data Collection

The dataset was collected using the yfinance library, which provides historical stock data directly from Yahoo Finance. The selected stock for this project is Google (GOOG), and the time range spans from:

- **Start Date:** 2012-01-01
- **End Date:** 2022-12-21

The dataset includes:

- Open price
- Close price
- High and Low prices
- Volume

The primary focus is on the **closing price**, as it reflects the final valuation of the stock for each trading day.

2.2 Data Preprocessing

After downloading the dataset:

- The index was reset to make the date a column.
- Missing values were removed using dropna().
- Only the **Close** column was selected for modeling.

The dataset was split into:

- **Training Data (80%)**
- **Testing Data (20%)**

This ensures that the model is trained on past data and evaluated on unseen data.

2.3 Exploratory Data Analysis (EDA)

To understand trends in the stock data, moving averages were calculated:

- **100-day Moving Average**
- **200-day Moving Average**

These were plotted alongside the actual closing prices to visualize trends and smoothing effects.

Observations:

- Moving averages help identify long-term trends.
- Crossovers between moving averages often indicate trend reversals.

Table 2. Classifier Results for NYSE.

	Normal Dataset				Leaked Dataset			
Classifier	**Precision**	**Recall**	**F Score**	**Accuracy**	**Precision**	**Recall**	**F Score**	**Accuracy**
Random Forest	0.61	0.58	0.6	0.52	0.78	0.95	0.86	0.82
Bagging	0.61	0.63	0.62	0.52	0.81	0.92	0.86	0.83
AdaBoost	0.62	0.64	0.63	0.54	0.68	0.77	0.72	0.66
Decision Trees	0.59	0.45	0.51	0.46	0.78	0.78	0.78	0.74
SVM	0.62	1	0.76	0.62	0.57	1	0.73	0.57
K-NN	0.64	0.49	0.55	0.51	0.68	0.75	0.71	0.65
ANN	0.63	0.51	0.57	0.51	0.6	0.8	0.69	0.59

2.4 Feature Scaling

Since neural networks perform better with normalized data:

- The closing price values were scaled using normalization techniques (e.g., MinMaxScaler).
- This ensures all values lie within a similar range, improving model convergence.

2.5 Model Architecture (LSTM)

The core of this project is the LSTM model, which is designed for sequence prediction.

Key characteristics:

- Captures temporal dependencies
- Maintains memory of previous inputs
- Handles vanishing gradient problems better than traditional RNNs

Typical architecture used:

- Multiple LSTM layers
- Dense (fully connected) output layer
- Activation functions for non-linearity

Input to the model:

- Sequences of previous stock prices (time steps)

Output:

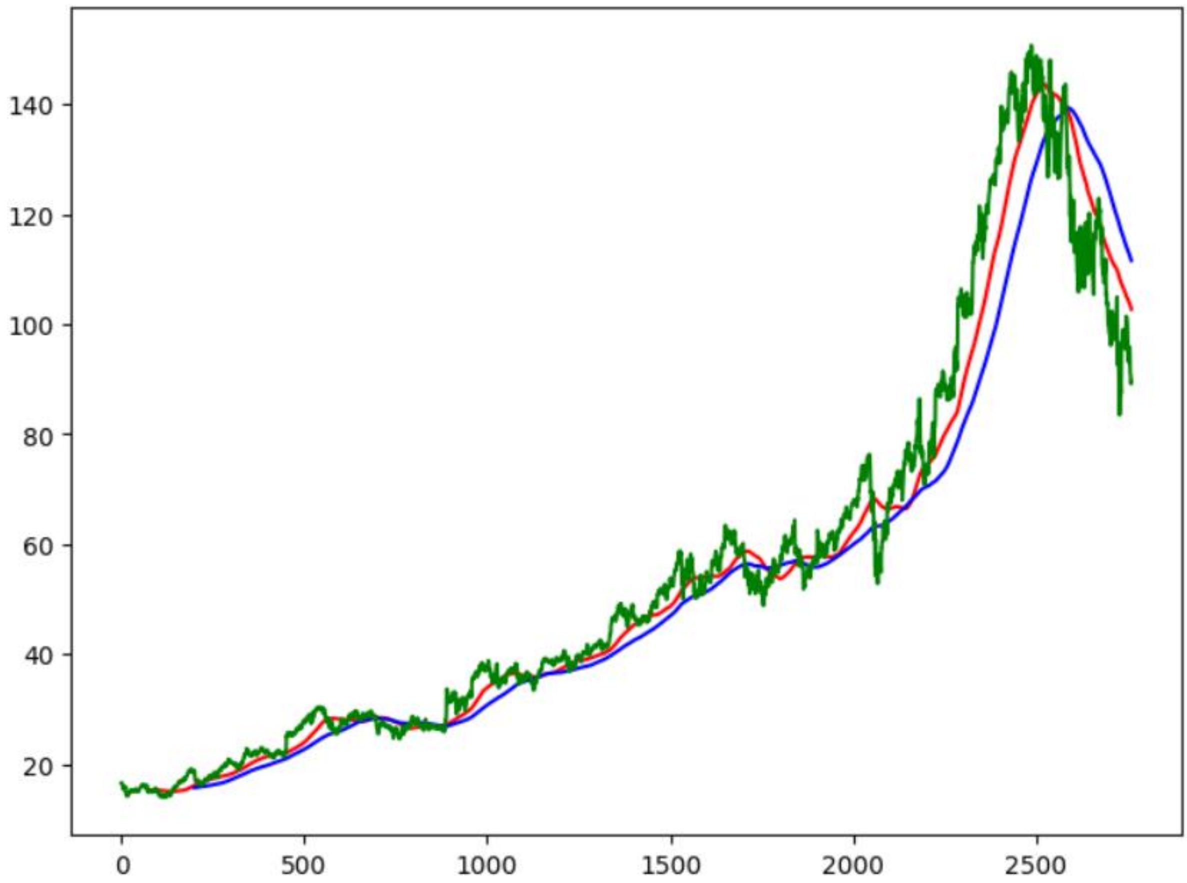
- Predicted next stock price

2.6 Training the Model

Steps involved:

1. Creating sequences from training data
2. Feeding sequences into the LSTM model
3. Training using backpropagation
4. Optimizing loss using an optimizer like Adam

- Loss function : Mean Squared Error (MSE)

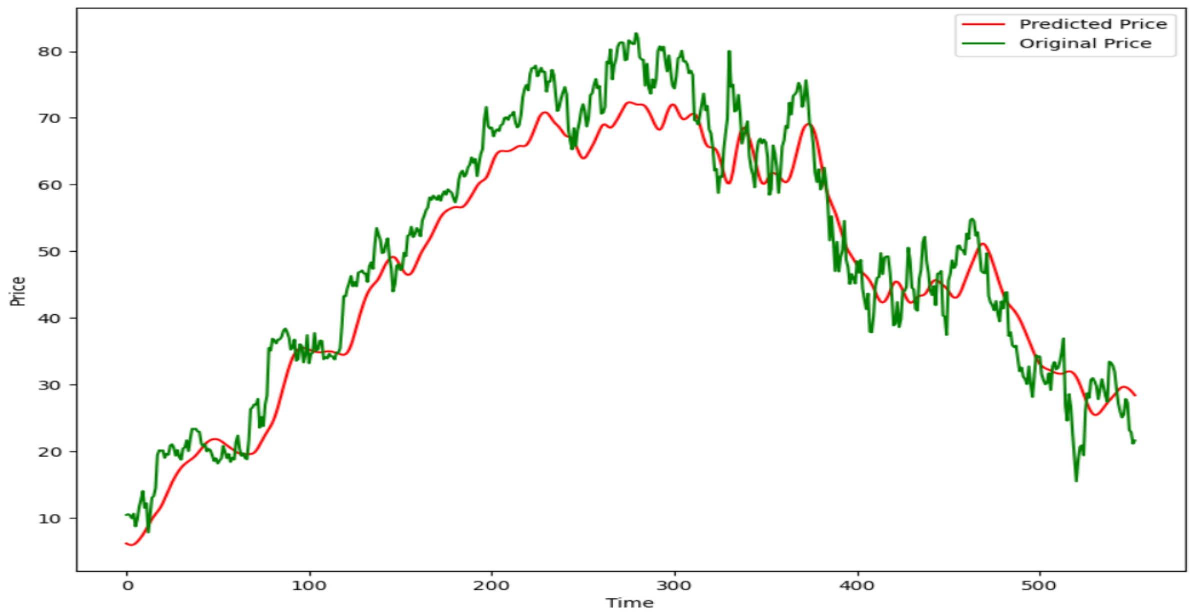


2.7 Model Evaluation

The model is evaluated using test data:

- Predictions are compared against actual stock prices
- Graphical comparison is used for visualization

Metrics considered:



- Prediction accuracy (visual)
- Error trends

3. Results

The results of the project show that:

- The model successfully learns general trends in stock prices.
- Predicted values closely follow the actual price curve, especially in stable regions.
- Slight deviations occur during highly volatile periods.

Key Observations:

- Moving averages align well with long-term trends.
- LSTM captures temporal patterns effectively.
- Predictions are smoother compared to real prices.

Visualization Insights:

- The predicted curve follows the actual trend but lags slightly.
- Sudden spikes or drops are harder to predict accurately.

---

4. Discussion

The results demonstrate the effectiveness of LSTM for time-series forecasting, but several important considerations arise:

Strengths:

- Handles sequential data efficiently
- Captures long-term dependencies
- Performs better than traditional regression models

Limitations:

- Cannot fully predict sudden market crashes or spikes
- Depends heavily on historical data
- Sensitive to hyperparameters

Challenges Faced:

- Data volatility

- Overfitting risk
- Need for proper scaling and preprocessing

Possible Improvements:

- Use Bidirectional LSTM (BiLSTM)
- Combine with CNN for feature extraction
- Incorporate external features like:
  - News sentiment
  - Economic indicators
- Hyperparameter tuning

---

5. Conclusion

In this project, we developed a stock price prediction model using LSTM neural networks. By leveraging historical stock data and deep learning techniques, we were able to capture patterns and trends in the data.

The model demonstrates that:

- Deep learning can be effectively applied to financial time series
- LSTM is well-suited for sequential prediction tasks
- Predictions are reasonably accurate for trend analysis

However, stock markets are influenced by many unpredictable external factors, making perfect prediction impossible. Therefore, such models should be used as **decision-support tools rather than absolute predictors**.

6. References

i. Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015). Predicting stock and stock price index movement using Trend Deterministic Data Preparation and machine learning techniques. Expert Systems with Applications, 42(1), 259–268. https://doi.org/10.1016/j.eswa.2014.07.040

ii. Pehlivanlı, A. Ç., Aşıkgil, B., & Gülay, G. (2016). Indicator selection with committee decision of filter methods for stock market price trend in ISE. Applied Soft Computing, 49, 792–800. https://doi.org/10.1016/j.asoc.2016.09.004 What is the

FTSE 100? | The Share Centre. (n.d.). Retrieved April 8, 2020, from https://www.share.com/a-guide-to-investing/before-youstart/what-is-the-ftse-100

iii. Witten, I. H., Frank, E., Hall, M. A., & Pal, C. J. (2016). Data Mining: Practical machine learning tools and techniques. Morgan Kaufmann.

iv. Zhang, X., Li, A., & Pan, R. (2016). Stock trend prediction based on a new status box method and AdaBoost probabilistic support vector machine. Applied Soft Computing, 49, 385–398. https://doi.org/10.1016/j.asoc.2016.08.026

v. Zhong, X., & Enke, D. (2017a). Forecasting daily stock market return using dimensionality reduction. Expert Systems with Applications, 67, 126– 139. https://doi.org/10.1016/j.eswa.2016.09.027

vi. Zhong, X., & Enke, D. (2017b). A comprehensive cluster and classification mining procedure for daily stock market return forecasting. Neurocomputing, 267, 152–168. https://doi.org/10.1016/j.neucom.2017.06.010

vii. Ballings, M., Van den Poel, D., Hespeels, N., & Gryp, R. (2015). Evaluating multiple classifiers for stock price direction prediction. Expert Systems with Applications, 42(20), 7046–7056. https://doi.org/10.1016/j.eswa.2015.05.013

viii. Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

ix. Chong, E., Han, C., & Park, F. C. (2017). Deep learning networks for stock market analysis and prediction: Methodology, data representations, and case studies. Expert Systems with Applications, 83, 187–205. https://doi.org/10.1016/j.eswa.2017.04.030

x. Cichosz, P. (2014). Data Mining Algorithms: Explained Using R. John Wiley & Sons

NAME :- RAM MOHAN HOTA , ASHISH SAMAL

SIC :- 23BCEH14 , 23BCED54

ROLL NO. :- 27 , 19

SEC :- CEN

SUBJECT :- ADVANCE MACHINE LEARNING